

Interpretable Geometric Analysis of the Humerus in Fat-Suppressed Shoulder MRI

Toward localization of rotator cuff tears via foundation-model segmentation
and direction-aware geometric features

Research Team
Universidad Nacional de Colombia

Topics in Computational Geometry — UNAL, Colombia

1st UEMS Congress 2026
Leuven, Belgium — 27–30 May 2026

Bridging computational geometry and musculoskeletal radiology

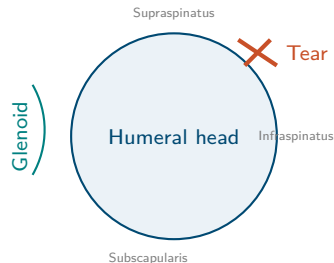
The clinical problem: rotator cuff tears

Why it matters

- Rotator cuff disease affects **>30% of adults over 60**, the leading cause of shoulder pain and disability.
- Tears arise from acute trauma or chronic degeneration (microtrauma + tendon-insertion ischemia).
- Accurate **localization** – which tendon, where along the insertion – drives surgical planning.

Diagnostic challenge

- MRI reading is expert-dependent and time-consuming.
- Inter-vendor variability (SIEMENS / GE / Philips) complicates automation.
- Equity gap: limited access to subspecialty radiology in low-resource settings

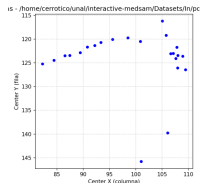
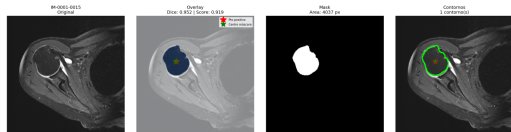


Axial schematic of the glenohumeral joint.
Tears disrupt the articular contour focally.

Imaging context: why fat-suppressed axial MRI?

Fat saturation (fs)

- Selective RF pulse **nulls fat signal** $\Rightarrow$ subcutaneous and marrow fat appear dark.
- Cortical bone, tendon, edema and joint fluid become **conspicuous**.
- Standard sequences here: **PD-fs** (anatomical detail + edema sensitivity) and **T1-fs**.



Why the axial plane?

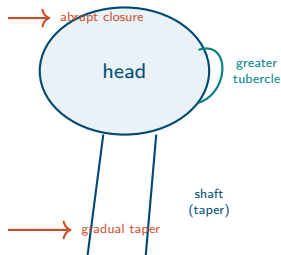
- The humeral head appears as **nested circles** that grow to a maximum (the “equator”) and shrink.
- Yields a clean *area-vs-slice* curve we can validate geometrically.
- Direct view of the glenohumeral contact surface where most tears manifest.

Axial PD-fs slice with humeral segmentation overlay
(*PixelSpacing* ≈ 0.47 mm; *matrix* 320×320)

From anatomy to geometry: two natural primitives

The humeral head $\approx$ spherical cap

- Articular cartilage covers a near-perfect spherical surface (radius $\sim 23\text{--}26$ mm in adults).
- Motivates a **global sphere fit** as a patient-specific anatomical reference.



The articular surface $\approx$ local circles

- In equatorial axial slices, the contact contour with the glenoid is locally circular.
- Motivates **per-slice circles** that *lie on* the articular surface (not approximate it).

Anatomy directly motivates two thresholds in the pipeline:
30% on the head side, **10%** on the shaft side.

Anatomical asymmetry

- **Head:** closes **abruptly** at the anatomical neck.

Hypothesis and objective

Hypothesis

A rotator cuff tear produces a **focal, geometrically detectable** discontinuity of the humeral articular contour. If we extract clinically interpretable geometric features from MRI, we can **localize** the lesion to a specific tendon insertion sector — without training a new neural network.

Objective

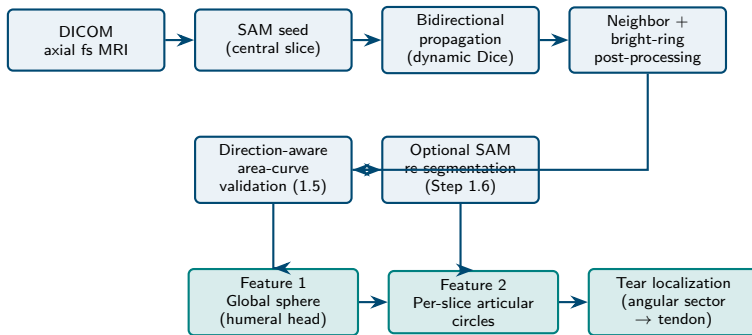
Build an **end-to-end pipeline** that:

- ① Segments the humerus across the entire axial volume from a **single seed slice**.
- ② Extracts two interpretable features: **(1)** a global humeral-head **sphere**; **(2)** per-slice articular **circles** that lie on the cartilage surface.
- ③ Uses these features as inputs for downstream tear-localization.

Design principles

- **Interpretable**: every output is geometrically meaningful and visually verifiable by a radiologist.
- **Vendor-agnostic**: works on SIEMENS, GE and Philips data without retraining.
- **Anatomy-aware**: pipeline parameters reflect humeral anatomy, not arbitrary heuristics.

Pipeline overview



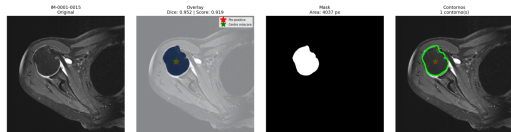
Foundation model (SAM, frozen) + classical geometry + clinical anatomy.

Each block produces auditable artifacts: masks, CSVs, validation reports, summary figures.

Stage 1: SAM segmentation + bidirectional propagation

Foundation-model segmentation

- **SAM (ViT-B)** as a frozen segmenter — no training, no labels.
- One seed slice (click *or* automatic point at $0.40W, 0.40H$).
- Morphological refinement: small-object removal, hole filling, opening/closing.

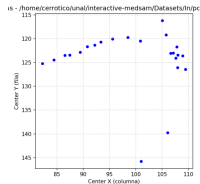
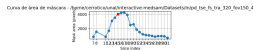


Bidirectional propagation, dynamic Dice

Centroid of slice $i \rightarrow$ positive prompt for $i \pm 1$. **Adaptive threshold:**

$$\tau_i = 0.45 + 0.25 \cdot |i - i_{\text{seed}}| / \Delta_{\text{max}}$$

Strict near the seed, permissive at extremes. **Negative-prompt** fallback for failures.



Per-slice segmentation overlay.

Green = mask boundary;

red dot = positive prompt centroid.

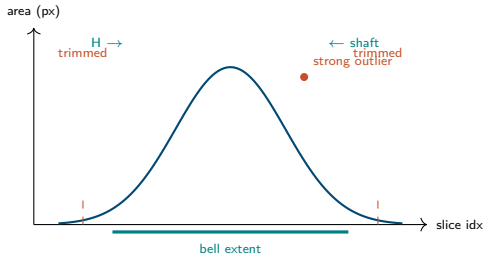
Stage 2: Direction-aware validation of the area curve

Reading anatomy from the DICOM header

- Compute slice normal $\mathbf{n} = \text{row} \times \text{col}$ from ImageOrientationPatient.
- Project ImagePositionPatient on $\mathbf{n} \Rightarrow$ **which end is the head.**
- Fallback: SliceLocation $\rightarrow$ InstanceNumber.

Asymmetric trim (anatomy-driven)

- Static floor from each border:
 - **30% of peak** on the head side (abrupt closure)
 - **10% of peak** on the shaft side (gradual taper)
- Reversal-from-peak: stop when the curve genuinely rises ($\geq 10\%$ head, $\geq 20\%$ shaft, ≥ 50 px absolute).
- Internal-outlier flagging: robust MAD residual



Schematic: per-slice mask area along the axial stack.
The pipeline trims tails and flags internal anomalies before fitting any geometry.

Feature 1: global humeral-head sphere

What it is

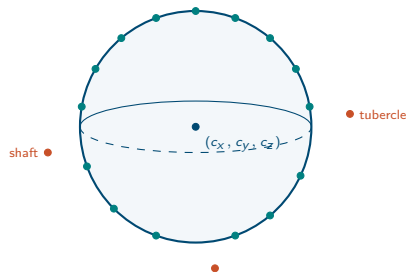
Single sphere fitted to the 3D contour cloud (DICOM millimeters): a patient-specific **anatomical reference** for the humeral head.

How it is computed

- Pixel $\rightarrow$ world via IPP, IOP, PixelSpacing.
- **RANSAC sphere**: 2000 iter., 4-point sampling, Kåsa fit.
- Inlier distance ≤ 2.0 mm; refit LSQ on inliers; sanity $r < 200$ mm.

Why it is clinically useful

- Patient-expected curvature (typical $r \in 23\text{--}26$ mm).
- Filters tubercles, shaft and artifacts automatically.
- Enables **normalized comparison** across patients and vendors.



Inliers (teal) define the humeral head; tubercles and shaft (orange) are correctly excluded.

Feature 2: per-slice articular circles

Key idea

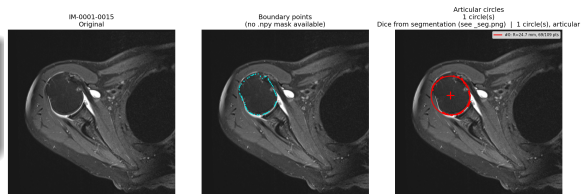
These circles do **not approximate** the sphere — they **lie on** the cartilage surface where the humeral head meets the glenoid.

How they are computed

- Articular slices: mask area $\geq 70\%$ of the peak.
- Project boundary to the slice's best-fit plane via **SVD**.
- Iterative 2D RANSAC + Kåsa fit (up to 3 circles/slice).
- Inlier ≤ 1.5 mm; ratio $\geq 35\%$; min 10 inliers.

What each circle stores

- Center (3D mm), radius, in-plane axes (**u**, **v**), normal.
- Inlier cloud + residual statistics.
- Re-projection on the original axial PNG for clinician review.



Five-panel render for one articular slice: original image, mask, contour, fitted circle, and overlay.

The fitted circle lies precisely on the articular contour.

The geometric trick: angular sectors $\rightarrow$ rotator cuff

Parameterizing the articular circle

Each per-slice circle has an in-plane basis $(\mathbf{u}, \mathbf{v})$ and a center $\mathbf{c}$. Any inlier $\mathbf{p}$ has angular coordinate

$$\theta = \text{atan2}((\mathbf{p} - \mathbf{c}) \cdot \mathbf{v}, (\mathbf{p} - \mathbf{c}) \cdot \mathbf{u}).$$

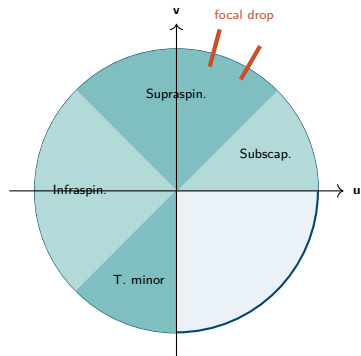
Anatomical sector mapping

Tendon insertions fall in **predictable angular sectors**:

- **Subscapularis** $\rightarrow$ lesser tub.: $\sim 0-45^\circ$.
- **Supraspinatus** $\rightarrow$ greater tub. (sup): $\sim 45-135^\circ$.
- **Infraspinatus** $\rightarrow$ greater tub. (post): $\sim 135-225^\circ$.
- **Teres minor** $\rightarrow$ greater tub. (inf-post): $\sim 225-270^\circ$.

Localization rule

A focal drop in inlier ratio in a specific sector $\rightarrow$ candidate tear in the corresponding tendon. at the



The articular circle, parameterized by θ , becomes a 1D *tendon map*.

Multi-vendor robustness and equity of access

Tested across vendors and sites

- **SIEMENS** – Quibdó, PD-fs TSE.
- **GE** – Optima MR360, Bogotá, PD-fs FSE.
- **Philips** – Ingenia, SOMA, ePDW SPAIR.
- Matrix 256×256 to 320×320 ; PD-fs and T1-fs.

Dataset	Kept	Articular
SIEMENS P5SE1 (Quibdó)	21/32	5
SIEMENS PD fs (fov150)	18/23	4
SIEMENS T1 fs #5	19/24	5
SIEMENS T1 fs #11	17/22	4
GE Optima MR360 (Bogotá)	16/21	4
Philips Ingenia (SOMA)	20/26	5

Why this generalizes

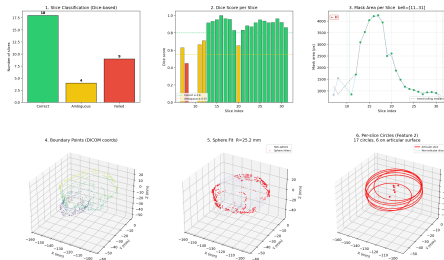
- Frozen SAM $\Rightarrow$ no domain-specific weights to overfit.
- Thresholds are **geometric** (mm, % of peak), not intensity-based.
- Direction handling reads DICOM headers, not vendor conventions.

Equity-of-access angle

Datasets include **Quibdó**, **Chocó**, a region with limited subspecialty radiology. Vendor-agnostic, interpretable AI directly supports diagnostic capacity in such settings.

Qualitative results: per-dataset summary

Humerus Boundary Analysis - ad tse fs tra 320 fow150.d



7. Analysis Summary

[illegible]

Summary (*pd_tse_fs_tra_320_fov150_4*): mask boundaries, area curve,
circle centers, reconstructed surface.

What the figure shows

- **Green**: slices kept for analysis.
- **Gray**: tails trimmed by direction-aware logic.
- **Red**: strong outliers excluded after Pass 2.
- Vertical lines: bell-extent boundaries.
- Caret $\text{H} \rightarrow$: which side of the volume is the head.

Per-slice circle visualization

Five-panel *_circle.png files let a radiologist verify each fit at a glance, slice by slice – **auditability by design**.

Limitations and clinical considerations

Current limitations

- **Validation cohort size:** 6 datasets across 3 vendors — proof of concept, not clinical validation.
- **Seed-slice selection:** automatic point at $(0.40W, 0.40H)$ assumes typical centering; pathological positioning may fail.
- **Pattern B without GPU:** when SAM grabs adjacent muscle/glenoid, weak outliers are flagged but not re-segmented unless GPU + SAM are available (Step 1.6).
- **Tear-localization rule is a hypothesis:** the angular-sector mapping needs prospective validation against arthroscopic ground truth.

What is required for clinical translation

- **Multi-center, larger cohort** with arthroscopy or surgical confirmation.
- Inter-reader study: do radiologists agree with the geometric flag location?
- Integration with PACS / DICOM-SR for routine workflow.
- Reproducibility: containerized pipeline, public benchmarks.

Risk profile

Because outputs are **geometric, visualized, and auditable**, the system fails “loudly” rather than silently — a key property for clinical deployment compared to opaque end-to-end models.

Conclusions and next steps

Take-home messages

- A **frozen foundation model** (SAM) plus **anatomy-aware classical geometry** produces interpretable shoulder MRI features without training.
- DICOM-header reading makes the pipeline **direction-aware** with asymmetric anatomy-driven thresholds.
- Two features (humeral sphere + per-slice articular circles) enable **tear localization by angular sector**.
- Robust across SIEMENS, GE, Philips; relevant for low-resource radiology settings.

Next steps

- Implement the angular-sector → tendon mapping end-to-end.
- Prospective validation against arthroscopy.
- Inter-reader and inter-vendor reproducibility study.
- Integration into a clinical reading viewer.

Acknowledgements

- Universidad Nacional de Colombia — Topics in Computational Geometry.
- Clínica Especialistas Reina Virgen María (Quibdó), Clínica Santa Ana (Bogotá), SOMA Radiology.
- Open-source: SAM (Meta AI), pydicom, scikit-image, OpenCV, NumPy/SciPy.

Thank you — questions welcome.

1st UEMS Congress 2026 — Leuven, Belgium